

BATTERY CHARGER And DISCHARGER

Using USB Power Supply

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Gadgets like electronic toys and musical players commonly use 1.2V rechargeable batteries in capacities ranging from 200mAh to 2800mAh, which need to be charged and maintained regularly.

Here we present a simple and low-cost device that can charge or discharge up to four 1.2V rechargeable batteries simultaneously with maximum current of around 100mA per battery. As the device uses USB power, it doesn't require any external power supply.

Circuit and working

Circuit of the battery charger and discharger with USB power supply is shown in Fig. 1. It is built around four 30-ohm, 1-watt resistors (R1 through R4), four 1N4007 diodes (D1 through D4), nine LEDs (LED1 through LED9), a TL431 shunt regulator (IC1), an LM324 quadruple op-amp (IC2), four 2.5V, 200mA small bulbs and a few other components.

Charger. All the four batteries (Batt.1 through Batt.4) are charged independently but simultaneously. The total charging current is below 500mA, which is the maximum current for many USB ports.

Many Ni-Cd chargers charge batteries only in series-connected pairs of two or four. The status of

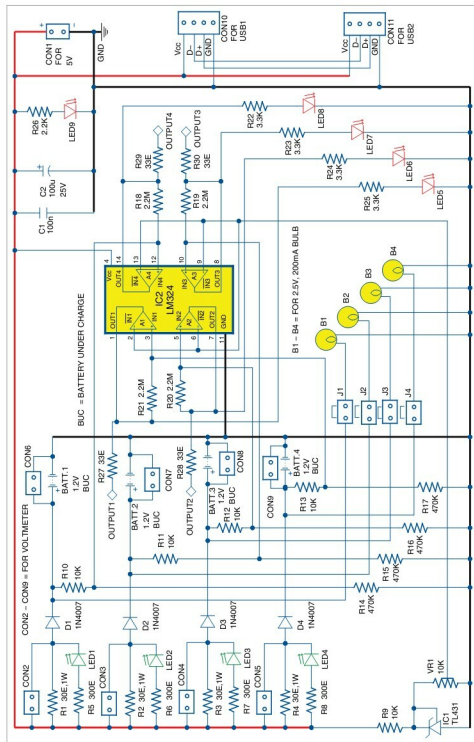


Fig. 1: Circuit of battery charger and discharger with USB power supply